Regression and Residuals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Regression and Residuals before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Train a Classifier

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Regression and Residuals trace and record the first step you can explain without help.

Use W2: Explain cancellation for **Regression** and **residual**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Regression and Residuals explanation against the supplied facts before discussing it.

Use the worked case for **Regression** and **residual**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Observed [6, 10, 13], predicted [8, 9, 11] cm. Give **residual**s, mean **residual** and error directions.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Regression and Residuals target to a trace and a justified conclusion.

Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual−predicted, calculate **residual** and absolute error. Interpret the sign.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Regression and Residuals case and show the deciding evidence.

Actual height=12 cm, **prediction**=9 cm. **Residual** means actual minus predicted.

Calculate and interpret.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Define **residual** as actual minus predicted. Actual height is 14 cm and **prediction** is 17 cm. Calculate **residual** and interpret its sign.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 19 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Regression and Residuals adds to Responsible Plant Classifier and keep the evidence.

Use a miniature growth-**prediction** side experiment to distinguish **regression evidence** from the main classifier metrics.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual−predicted, calculate **residual** and absolute error. Interpret the sign.

> **My independently reasoned answer**
>
> _____
>
> _____
>
> _____
>
> _____

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Calculate residuals

> Calculate residuals. Apply the supplied Regression and Residuals facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Regression and Residuals", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Compute the worked **residual**s and identify **overprediction**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Explain cancellation

> Explain cancellation. Apply the supplied Regression and Residuals facts and keep your reasoning.

Why does mean **residual** zero not imply perfect **prediction**s?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Read a pattern

Read a pattern. Apply the supplied Regression and Residuals facts and keep your reasoning.

Interpret **residual**s -2, 0, 3 ordered by growth stage.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied

> Regression and Residuals facts and keep your reasoning.

Observed [6, 10, 13], predicted [8, 9, 11] cm. Give **residual**s, mean **residual** and error directions.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Check units

> Check units. Apply the supplied Regression and Residuals facts and keep your reasoning.

A report combines 2 cm and .02 m as if they were different magnitudes. Repair it.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Regression and Residuals

> Use residual signs to say whether each plant-height estimate is too high or low. Explain the labelled step before checking the changed condition.

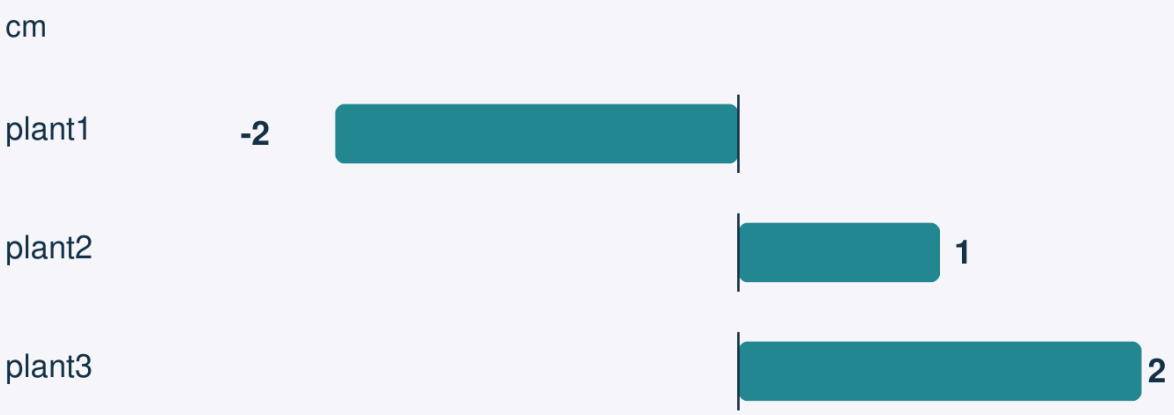
Use **residual** signs to say whether each plant-height estimate is too high or low.

Residual=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/19

Observed minus predicted height



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/19

Regression and Residuals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Subtract matched pairs

Residuals are5-7=-2, 8-7=1, 12-10=2 cm.

2. Read directions

First prediction is too high by2; second too low by1; third too low by2.

3. Avoid cancellation

Mean residual is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

CHANGE ONE THING

Raise all predictions by1 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use **residual** signs to say whether each plant-height estimate is too high or low.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

1. Read directions

2. Avoid cancellation

3. Subtract matched pairs

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Avoid cancellation Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Read the picture in order.

Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Raise all **prediction**s by1 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Each **residual** decreases by1 under the observed-minus-predicted convention.

Repair a missing building block

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

Explain each Regression and Residuals picture from its labels, then record a relevant limit.

01. Regression and Residuals: the case

AI ACADEMY / AI-4/19 · 01 · LOOK AT THE FACTS

Regression and Residuals: the case

The situation

Use residual signs to say whether each plant-height estimate is too high or low.

Supplied evidence

Residual=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain regression and residuals using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

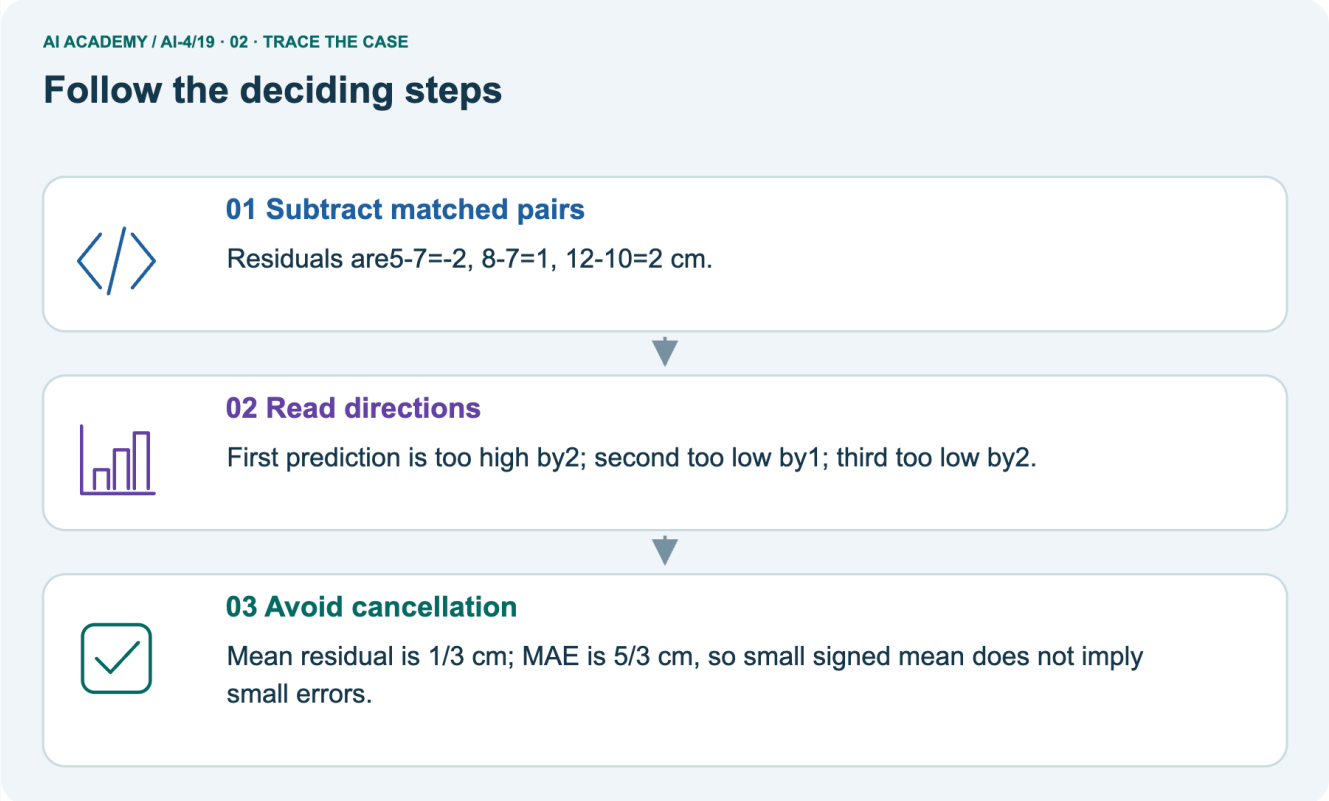
Use **residual** signs to say whether each plant-height estimate is too high or low. Supplied facts: **Residual**=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Subtract matched pairs: **Residual**s are5-7=-2, 8-7=1, 12-10=2 cm. → Read directions: First **prediction** is too high by2; second too low by1; third too low by2. → Avoid cancellation: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Observed minus predicted height



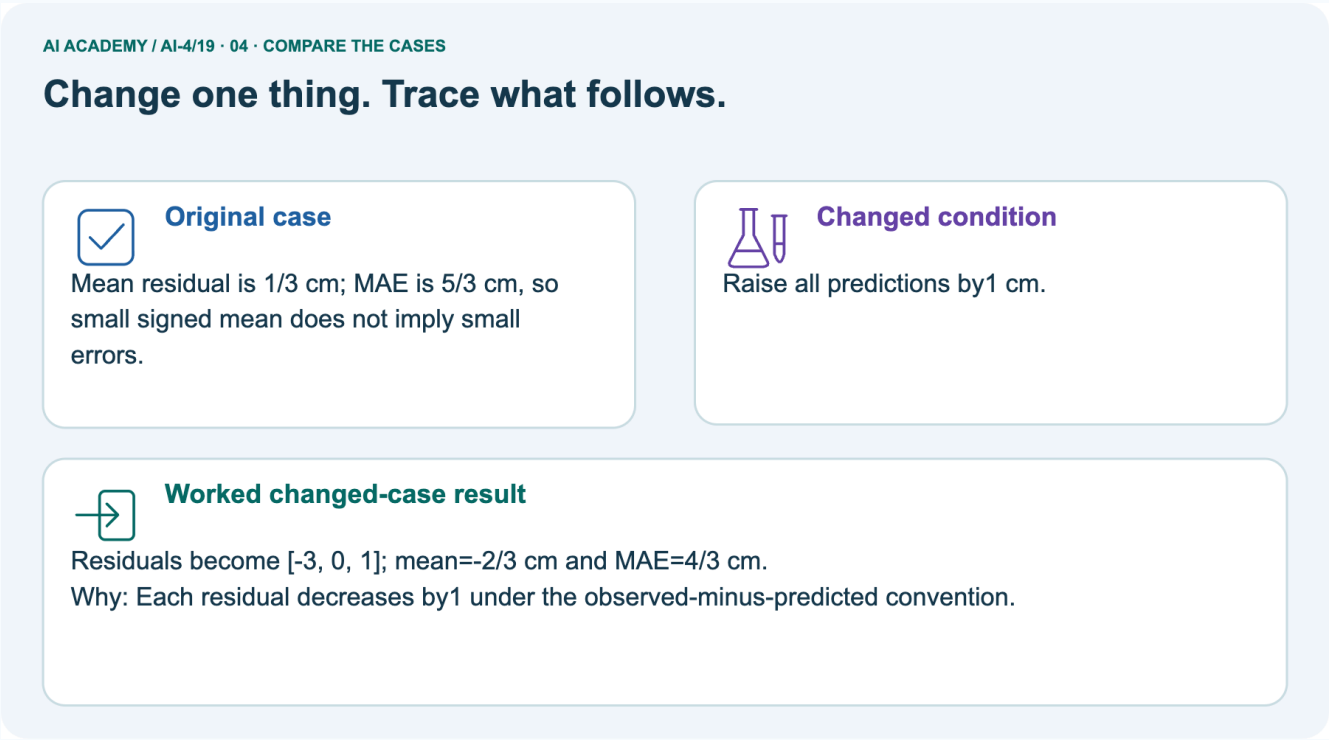
Follow the labelled picture with the matching explanation. Open the vector diagram

Observed minus predicted height. Units: cm. plant1: -2. plant2: 1. plant3: 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

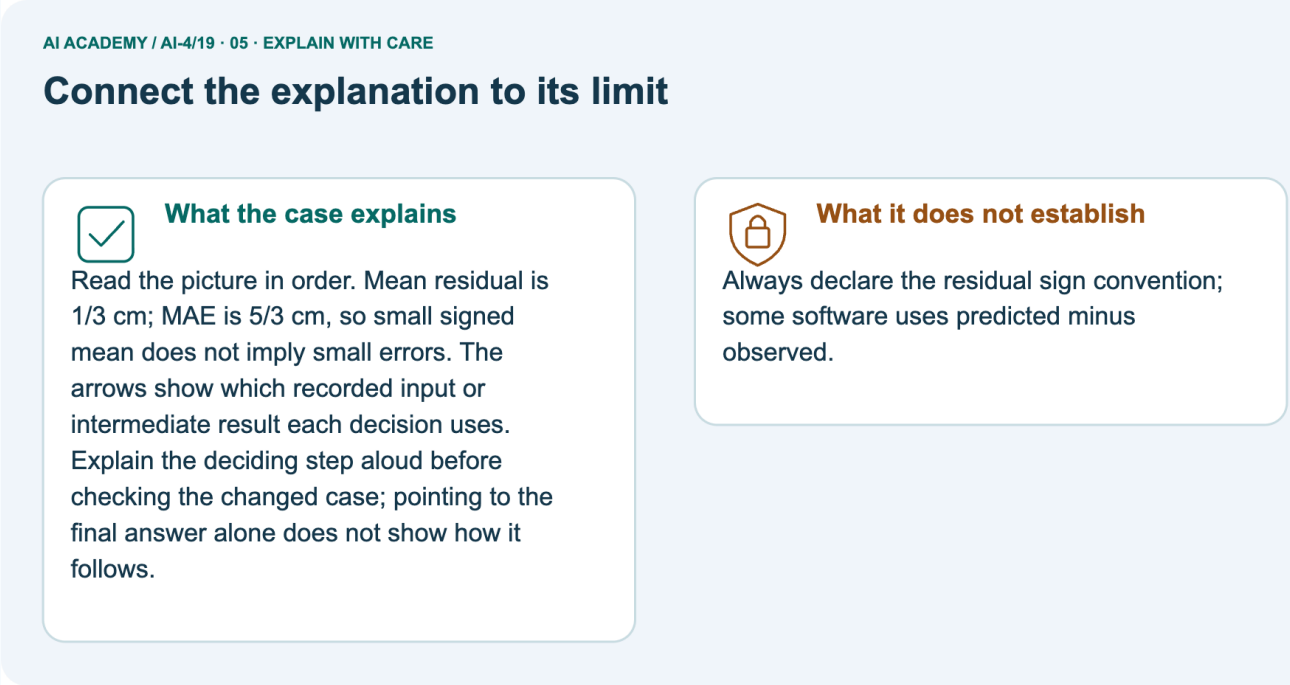
Original case: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Changed condition: Raise all **prediction**s by1 cm.

Expected result: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Always declare the **residual** sign convention; some software uses predicted minus observed.

> Explain the case, then point to a limit on the claim.

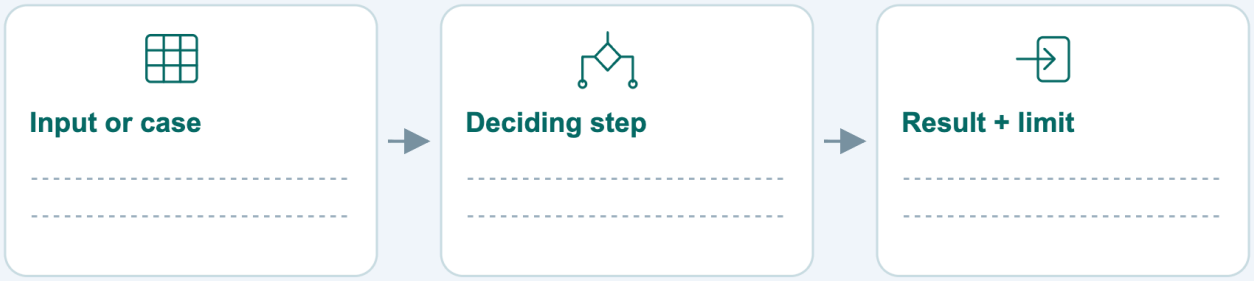
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-4/19 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.